



Heart Disease Prediction: Logistic Regression



Jared Maksoud, Danny Mendoza, Aren Mizuno, Kirk Waller

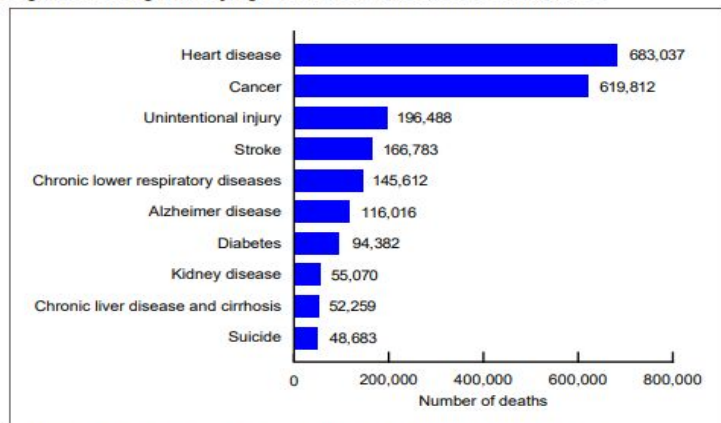
Introduction

What health and demographic factors impact a patient's likelihood of having heart disease?

We aim to predict the presence of heart disease using routinely collected clinical variables to enable early identification of high-risk patients.



Figure. Leading underlying causes of death: United States, 2024



NOTES: Deaths that occurred in the United States among residents of U.S. territories and foreign countries were excluded. Estimates for 2024 are based on provisional data as of June 1, 2025 (provisional mortality data are available from <https://wonder.cdc.gov/>). Provisional data are incomplete, and data from December are less complete because of reporting lags. SOURCE: National Center for Health Statistics, National Vital Statistics System, mortality.

Heart disease was the leading cause of death in the United States in 2024

- CDC & National Center for Health Statistics

Our model can help reduce mortality, lower health care costs, and detect health factors early

Data Description

The dataset comes from Kaggle and combines heart data from various hospitals around the world. It contains **918 patient records** and 12 original columns, expanded via one-hot encoding.

Target Variable

HeartDisease: Binary (1 = Disease, 0 = No Disease)

Age: Patient age in years

Sex: Sex of patient (M/F)

RestingBP: Resting blood pressure (mmHg)

Cholesterol: Serum cholesterol (mg/dl)

FastingBS: Fasting blood sugar >120 mg/dl (1/0)

MaxHR: Maximum heart rate achieved

ExerciseAngina: Exercise-induced angina (Y/N)

Oldpeak: ST-segment depression from exercise

ST_Slope: Slope of peak exercise ST segment

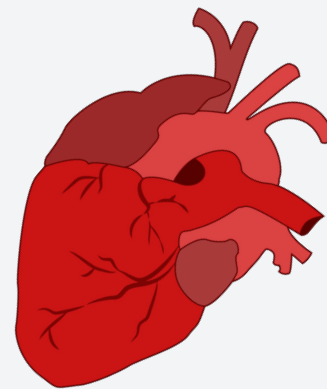
ChestPainType: Type of chest pain (ATA, NAP, etc.)

RestingECG: Resting electrocardiogram results (Normal, ST, LVH)

Data Description

- Categorical Data was encoded using dummy variables
- Description of numerical data:

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	HeartDisease
count	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.396514	198.799564	0.233115	136.809368	0.887364	0.553377
std	9.432617	18.514154	109.384145	0.423046	25.460334	1.066570	0.497414
min	28.000000	0.000000	0.000000	0.000000	60.000000	-2.600000	0.000000
25%	47.000000	120.000000	173.250000	0.000000	120.000000	0.000000	0.000000
50%	54.000000	130.000000	223.000000	0.000000	138.000000	0.600000	1.000000
75%	60.000000	140.000000	267.000000	0.000000	156.000000	1.500000	1.000000
max	77.000000	200.000000	603.000000	1.000000	202.000000	6.200000	1.000000



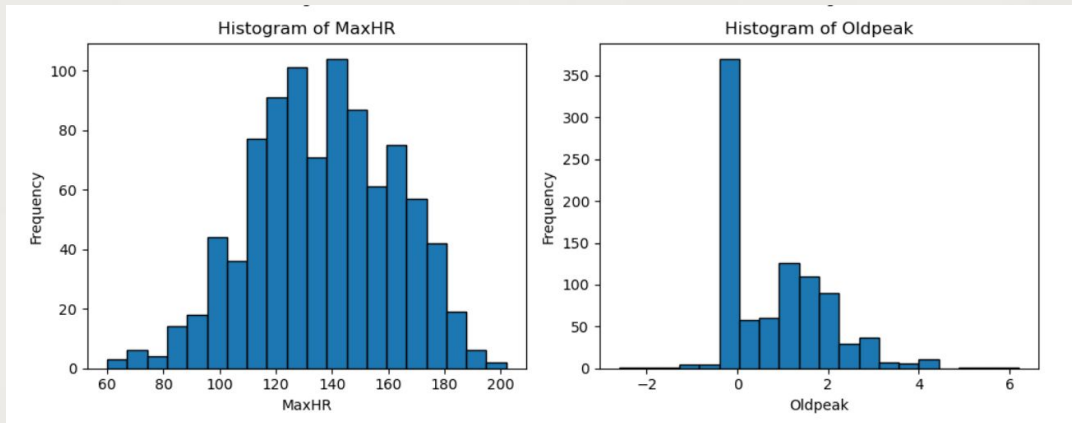
- Mainly an older population
- Mean of HeartDisease is closer to 1 → class imbalance
- Oldpeak has negative values → data anomaly

Data Description

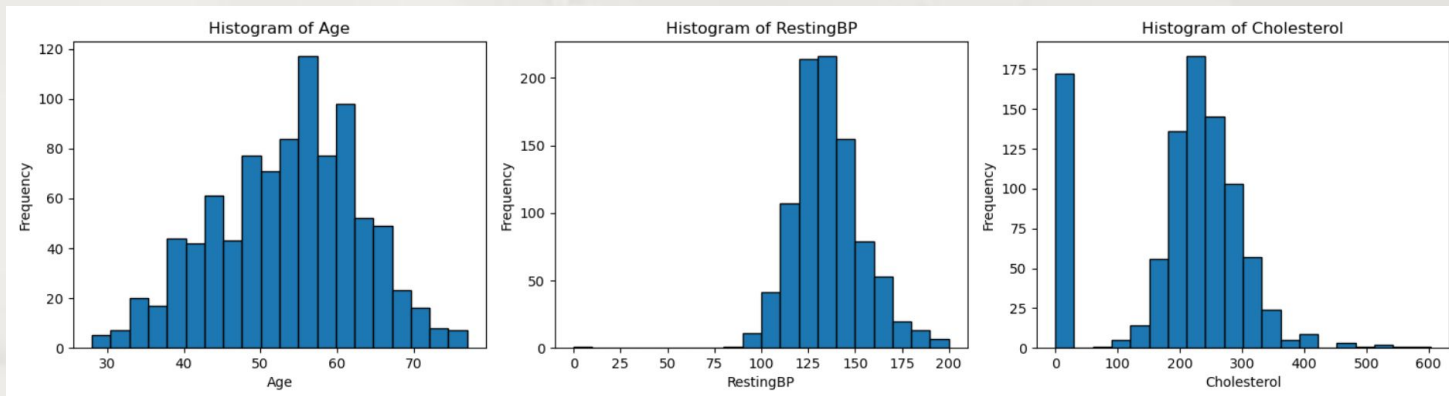
Total Entries: 918

Total Columns: 12

Total Numerical Features: 5



- Filtered out any negative values in Oldpeak and then applied log transformation to Oldpeak due to right-skewed distribution



- IQR outlier detection automatically filtered out rows with 0 included in Cholesterol column

Data Description

Final Dataset:

- 700 observations
- Ready for modeling



	Age	Sex	RestingBP	Cholesterol	FastingBS	MaxHR	ExerciseAngina	Oldpeak	HeartDisease	ChestPainType_ATA
count	700.000000	700.000000	700.000000	700.000000	700.000000	700.000000	700.000000	700.000000	700.000000	700.000000
mean	52.748571	0.760000	131.557143	239.587143	0.162857	140.555714	0.374286	0.830571	0.462857	0.228571
std	9.525341	0.427389	15.435718	50.286556	0.369499	24.383231	0.484284	0.959439	0.498975	0.420213
min	28.000000	0.000000	92.000000	85.000000	0.000000	71.000000	0.000000	0.000000	0.000000	0.000000
25%	46.000000	1.000000	120.000000	206.000000	0.000000	122.000000	0.000000	0.000000	0.000000	0.000000
50%	54.000000	1.000000	130.000000	235.000000	0.000000	140.000000	0.000000	0.400000	0.000000	0.000000
75%	59.000000	1.000000	140.000000	272.000000	0.000000	160.000000	1.000000	1.500000	1.000000	0.000000
max	77.000000	1.000000	170.000000	394.000000	1.000000	202.000000	1.000000	3.600000	1.000000	1.000000

*Notebook shows the description of the entire dataset

Model Specification

- Logistic Regression
 - Our target variable, HeartDisease, is binary: 1 = Disease, 0 = No Disease
 - Logistic Regression is a Generalized Linear Model
 - Predicts probability of a binary outcome
- Model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

- Link function: logit transforms the probability of HeartDisease into log odds
- p : Probability that $y = 1$, or that patient has heart disease
- $x_1, x_2, \dots, x_k$: predictor variables
- β_0 : intercept
- $\beta_1, \beta_2, \dots, \beta_k$: coefficients of predictors

Model Specification

Maximum Likelihood Estimation (MLE) Method & Key Equations

- Outcome variable: (HeartDisease $\in \{0,1\}$) $\rightarrow$ modeled as Bernoulli
- Bernoulli Distribution:

$$f(y; p) = p^y \cdot (1 - p)^{1-y}$$

- Likelihood Function:

$$L(y; \theta) = \prod_{i=1}^n p^{y_i} \cdot (1 - p)^{1-y_i}$$

- Log-likelihood Function

$$l(y; \theta) = \log p \cdot \sum_{i=1}^n y_i + \log(1 - p) \cdot (n - \sum_{i=1}^n y_i)$$

- Solution after taking FOCs:

$$\hat{p} = \frac{\sum_{i=1}^n y_i}{n}$$

Variable Selection

Initial Model: Significance Check

- Fit the full model with all candidate predictors and evaluated significance at the 5% level
- We feature engineered interaction terms with Sex and numerical variables because we wanted to test whether sex modifies the effect of other predictors
- All feature-engineered interaction variables were insignificant

Significant: p-value < 0.05

Variable	p-value
const	0.004
Sex	0.041
ExerciseAngina	0.000
ChestPainType_ATA	0.000
ChestPainType_NAP	0.000
ChestPainType_TA	0.009
ST_Slope_Up	0.000

Insignificant: p-value > 0.05

Variable	p-value
Age	0.785
RestingBP	0.137
Cholesterol	0.084
FastingBS	0.618
MaxHR	0.893
RestingECG_LVH	0.913
RestingECG_ST	0.533
ST_Slope_Down	0.141
log_oldpeak	0.162
Age_Sex_inter	0.727
RestingBP_Sex_inter	0.174
Cholesterol_Sex_inter	0.111
MaxHR_Sex_inter	0.779
Cholesterol_FastingBS_inter	0.505

Variable Selection

Regularization

Optimal penalty strength λ was selected via **10-fold cross-validation** using the **AOC-ROC** metric

LASSO (L1 Penalty)

Encourages sparsity by shrinking some coefficients to exactly zero, performing feature selection.

$$Loss = \sum (y_i - \hat{y}_i)^2 + \lambda |\beta_j|$$

$$\lambda = 10$$

Ridge (L2 Penalty)

Shrinks coefficients towards zero but does not set them to zero, handling multicollinearity.

$$Loss = \sum (y_i - \hat{y}_i)^2 + \lambda \beta_j^2$$

$$\lambda = 0.1$$

Elastic Net ($\alpha = 0.5$)

A hybrid approach that combines the penalties of LASSO and Ridge, balancing feature selection and shrinkage.

$$Loss = \sum (y_i - \hat{y}_i)^2 + \lambda_1 |\beta_j| + \lambda_2 \beta_j^2$$

$$\lambda = 0.1, 0.1$$

Variable Selection

Regularization

Features with Absolute Value of Shrinkage ≥ 0.25

Feature	Statsmodels Coefficient	LASSO Drop	Ridge Drop	Elastic Net Drop
Sex	4.041135	0.731002	3.801953	3.803266
RestingBP_Sex_inter	-2.079468	0.484346	2.193751	2.193202
Cholesterol_Sex_inter	-1.277936	0.130467	1.287616	1.287327
MaxHR_Sex_inter	-0.290018	0.162723	0.409552	0.371502
Cholesterol_FastingBS_inter	0.504760	0.120391	0.403687	0.404025
ST_Slope_Up	-1.308922	0.008968	0.272361	0.269920

- LASSO shrinks no variables completely to 0
- Almost all interaction terms were heavily shrunk
- All heavily-shrunk variables except Sex and ST_Slope_Up were statistically insignificant in the unregularized model
- Sex shrinks heavily because many strong interaction terms with Sex absorb its effect

Variable Selection

Multicollinearity – VIF Analysis

Features with $VIF \geq 5$

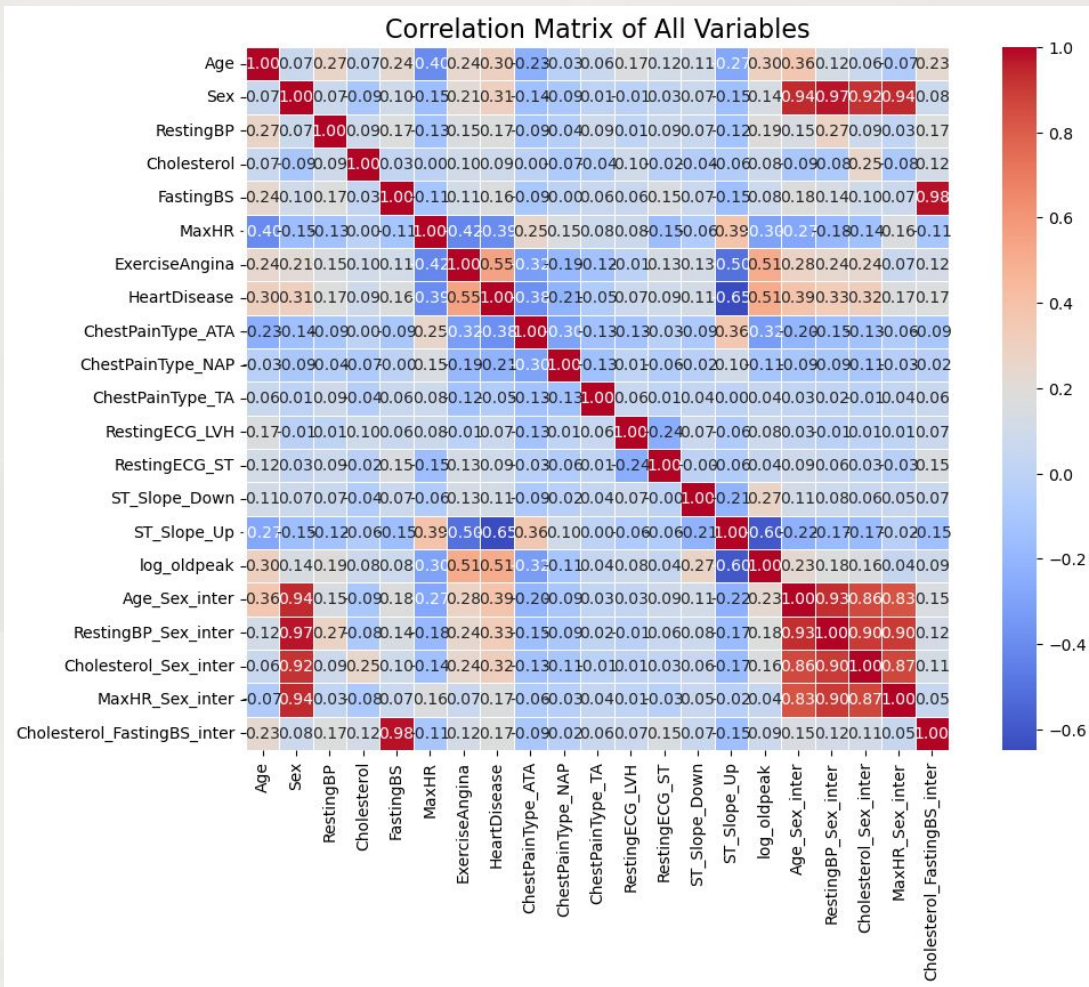
Feature	VIF
const	622.710123
Age	5.243968
Sex	162.238715
FastingBS	26.780288
MaxHR	5.208618
Age_Sex_inter	44.839234
RestingBP_Sex_inter	86.926491
Cholesterol_Sex_inter	26.811214
MaxHR_Sex_inter	46.311243
Cholesterol_FastingBS_inter	26.951717

- High VIF values indicate severe multicollinearity
- 5 is the conventional threshold
- All variables except Sex with $VIF \geq 5$ were statistically insignificant in the full regression
- All interaction terms have high VIF
- Sex has a high VIF as it is entangled in many interaction terms
- RestingBP and Cholesterol were close to 5 and are also statistically insignificant

Variable Selection

Matrix

- Matrix supports VIF numbers
- Interaction terms are highly correlated with their corresponding main variables and with each other



Variable Selection

Final Features

- We selected only the variables that remained statistically significant in the full model:
 - *Sex*
 - *ExerciseAngina*
 - *ChestPainType_ATA*
 - *ChestPainType_NAP*
 - *ChestPainType_TA*
 - *ST_Slope_Up*

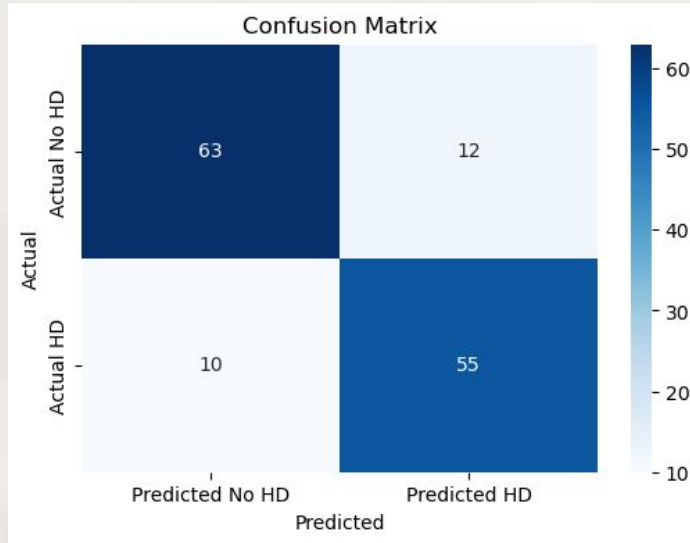
Likelihood Ratio Test:

- Evaluates difference in log-likelihoods and tests whether that improvement is large enough to be unlikely by chance
- LRT between the full model (including all predictors) and the reduced model (containing only statistically significant variables)
- P-value: 0.24
- Fail to reject the Null Hypothesis that the reduced model fits the data just as well as the full model
- The simpler, final model is sufficient

Results and Interpretation

Evaluation

Metric	Value	Equation
Accuracy	0.8429	$(tp+tn)/(tp+tn+fp+fn)$
Precision	0.8209	$tp/(tp+fp)$
Recall (Sensitivity)	0.8462	$tp/(tp+fn)$
Specificity	0.8400	$tn/(tn+fp)$
F1 Score	0.8333	$2/(1/precision+1/recall)$



- Correctly classified 84% of all cases
- Recall – model successfully identifies most patients who truly have heart disease
- Specificity – model avoids mislabeling healthy individuals as diseased
- The balanced F1 score reflects a strong trade-off between catching true cases and avoiding false alarms
- Only 12 FP and 10 FN
- All VIF lower than 5

Results and Interpretation

Coefficients and Significance

Variable	Coefficient	Std. Error	z-Value	p-Value
const	-0.4147	0.143	-2.904	0.004
Sex	0.8003	0.152	5.273	0.000
ExerciseAngina	0.6487	0.139	4.662	0.000
ChestPainType_ATA	-0.8532	0.172	-4.951	0.000
ChestPainType_NAP	-0.6530	0.139	-4.687	0.000
ChestPainType_TA	-0.2868	0.121	-2.375	0.018
ST_Slope_Up	-1.3983	0.146	-9.599	0.000

- All variables are statistically significant at 5% level
- Sex and ExerciseAngina increase the log odds of heart disease
- ChestPainType ATA, NAP, and TA all decrease log-odds relative to the asymptomatic (ASY) baseline — surprising because asymptomatic patients typically have lower risk
- ST_Slope_Up shows a strong negative coefficient, indicating a protective pattern

Sensitivity Analysis

Sex:

Being male **more than doubles** the odds of heart disease.

Exercise Angina:

Exercise induced angina **nearly doubles** the odds of heart disease.

Chest Pain Type ATA:

ATA chest pain reduces odds by **57%** compared to the asymptomatic baseline.

Chest Pain Type NAP:

NAP chest pain reduces odds by **48%**.

Chest Pain Type TA:

TA chest pain reduces odds by **25%**.

Variable	Odds Ratio (e^{β})
Sex	$e^{0.8003} = 2.23$
Exercise Angina	$e^{0.6487} = 1.91$
Chest Pain Type: ATA	$e^{-0.8532} = 0.43$
Chest Pain Type: NAP	$e^{-0.6530} = 0.52$
Chest Pain Type: TA	$e^{-0.2868} = 0.75$
ST_Slope_Up	$e^{-1.3983} = 0.25$

ST_Slope_Up:

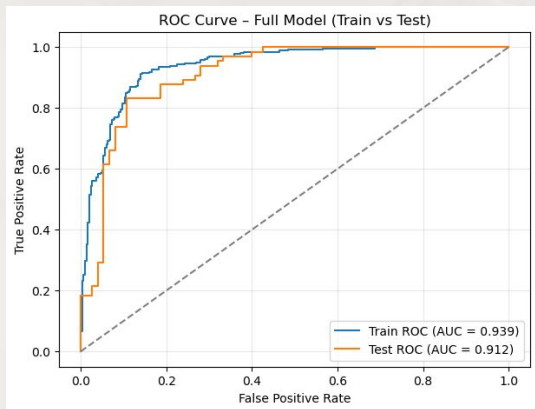
Patients with an upsloping ST segment during peak exercise have **75%** lower odds of having heart disease compared to patients with a flat ST segment.

Prediction (Out-of-sample)

Original Model AUC Curve

AUC-ROC:

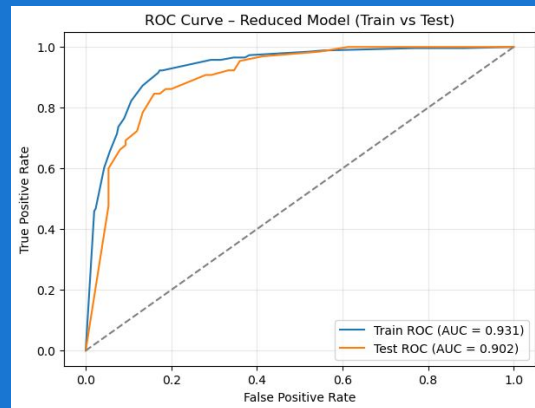
- High Training and Testing AUC
- Higher predictive power



Reduced Model AUC Curve

AUC-ROC:

- Decreased Training and Testing AUC
- Worse at generalization



Model Limitations

- Data is a cross-sectional sample with no duplicates, all entries are independent
- We removed 23.75% of the data which we considered outliers
- Severe multicollinearity in original data
- Large class imbalance between people with heart disease vs people without



Limited Data

Data may not generalize to populations with different prevalence or risk-factor distributions.



External Validation

Validation on external cohorts is recommended before clinical deployment.

Conclusions

Key Takeaways

- Logistic regression produced a strong, interpretable heart-disease prediction model
- **Sex, Exercise Angina, Chest Pain Type, and ST_Slope** emerged as the most influential predictors.
- Sensitivity analysis showed large, directional shifts in risk, consistent with clinical expectations for most features.

Practical Relevance

- Results highlight how simple, routinely collected clinical features can provide meaningful early risk stratification.
- Model interpretability makes it suitable for clinical decision support, patient screening, or educational use.

Possible Extensions

- Incorporate more advanced feature engineering as well as non-linear models like random forest or gradient boosting
- Validate performance on external or real clinical datasets



THANK YOU